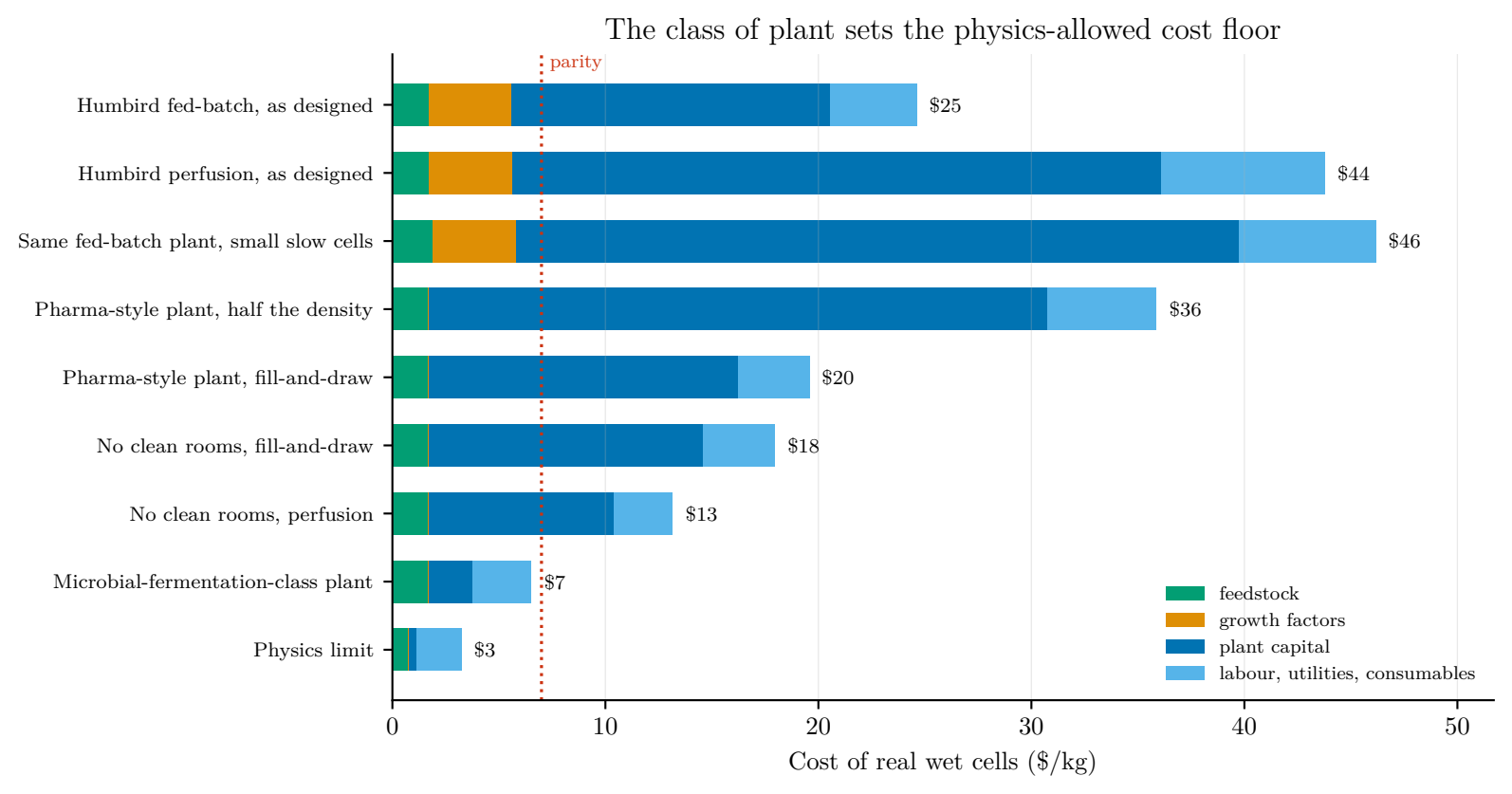

The class of plant sets the physics-allowed cost floor
parity
Humbird fed-batch, as designed
Humbird perfusion, as designed
Same fed-batch plant, small slow cells
Pharma-style plant, half the density
Pharma-style plant, fill-and-draw
No clean rooms, fill-and-draw
No clean rooms, perfusion
Microbial-fermentation-class plant
Physics limit
$25
$44
$46
$36
$20
$18
$13
$7
$3
feedstock
growth factors
plant capital
labour, utilities, consumables
0
10
20
30
40
50
Cost of real wet cells ($/kg)